

◆ Course Information

- This course introduces participants to practical AI and Industry 4.0 applications for manufacturing competitiveness. Participants will learn what AI, IoT and smart factory concepts mean on the factory floor, assess their current readiness, identify realistic digital opportunities and build a phased transformation roadmap that fits their operation.

◆ Course Objectives

- Upon completion of this program, participants will be able to:
- Understand practical AI, IoT and Industry 4.0 concepts for manufacturing
- Identify real factory-floor use cases in production, quality, maintenance and planning
- Assess current digital maturity, data readiness and implementation gaps
- Prioritise AI and IoT opportunities based on impact, feasibility and risk
- Build a pragmatic phased digital transformation roadmap for their organisation

◆ Who Should Attend

This course is designed for Management, Managers, Team Leaders, Engineers, Supervisors and teams involved in manufacturing competitiveness, digitalisation and operational improvement.

Included but not limited to: Production, Engineering, Maintenance, Quality Assurance/Quality Control, Planning, IT, Automation, Process Improvement, Continuous Improvement, Lean Operations and Industry 4.0 project teams.

◆ Methodology

Variety of training techniques and methods: including actual customized case study for real-life application
Lively two-way interaction training facilitation.

Role play to give feedback to subordinates and group presentation to dive deeper understanding of the techniques and application in the real world.

◆ Course Schedule

8:30	Registration	14:00	Training Resumes
9:00	Training Begins	15:30	Afternoon Break
10:30	Break	15:45	Training Resumes
10:50	Training Resumes	17:00	Training Concludes
12:50	Lunch		

Applicable for both days



Course Content

Day 1

Module 1:

AI & Industry 4.0 Fundamentals

What AI, IoT, automation and smart factory mean in manufacturing
Difference between hype and practical use cases

Why Industry 4.0 matters for competitiveness

Module 2:

Factory-Floor AI & IoT Use Cases

AI for production visibility and decision support

IoT data collection and equipment connectivity

Quality inspection, downtime monitoring and predictive maintenance examples

Module 3:

Industry 4.0 Readiness Assessment

Current digital maturity and data availability

Process stability, KPI discipline and system readiness

People, leadership and management readiness

Module 4:

Opportunity Identification

Identify operational pain points and digital gaps

Prioritise opportunities by impact and feasibility

Select realistic quick wins and pilot areas

Day 1 Continued

Module 5:

Digital Transformation Roadmap

Build a phased implementation approach

Pilot project, data requirement and ownership

Avoid common AI and Industry 4.0 implementation mistakes

Module 6:

Practical Workshop

Assess own organisation readiness

Map possible AI or IoT use cases

Prepare a simple transformation roadmap

Module 7:

Wrap-up

Key learning recap

Action plan for next steps

Questions

Feedback session